



Paper Number 5

Improvement of plant nutrient management for better farmer livelihood, food security and environment in Sri Lanka*

* This country report has not been formally edited and the designations and terminology used are those of the author.

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Summary

Sri Lanka is an agricultural country with diverse agro-climate and land resources. Agricultural soils are laterites which are generally acidic, low in organic matter and would require sound phosphorous management. Aside from low per capita land availability, soil fertility generally limits crop production and would require considerable attention and support. However, fertilizers necessary for improving yields of food crops are imported. The total quantity of fertilizer imported in 2002 was 568 072 mt. About 59 percent of this importation is Urea and the rest are divided into other fertilizers such as sulphate of ammonia, triplesuperphosphate and muriate of potash. Nutrient use imbalance and the general decline in soil fertility in Sri Lanka is primarily policy-induced. The national government adopted a subsidy scheme for Urea, which inadvertently resulted in increased Urea consumption and the evident reduction in the use of P and K fertilizers. To alleviate the situation, the national government has taken steps to locally produce phosphate fertilizers using local rock phosphate (eppawala rock phosphate). In addition, recently, to help farmers in adopting the rational use of Urea and general balance use of available organic and inorganic fertilizers the Ministry of Agriculture and the Department of Agriculture developed a national policy for the promotion of Integrated Plant Nutrition System (IPNS) and its adoption by the farmers in Sri Lanka.

1. Introduction

Sri Lanka is an island which is situated in the Indian Ocean and lies between latitude 5°55' and 9°50'N and longitude 79°42' and 81°53'E. The total land area of the island is 65 610 sq km. Population is estimated to be 20 million with an annual growth rate of 1.2 percent. The contribution to the gross domestic product (GDP) by agriculture is estimated to be 18.6 percent and employment in the agriculture sector is 37.7 percent. Contribution to the GDP in the agriculture sector by rice is 13.8 percent and other food crops 33.8 percent. Sri Lanka is divided into three major zones on the basis of annual rainfall viz. wet (>2 000 mm), intermediate (1 000-2 000 mm) and dry (<1 000 mm) zones. The wet and intermediate zones are further categorized into low, mid and up-country regions based on elevation. The low country is situated at elevation between 0 and 300 m above mean sea level (MSL), while midcountry is considered as the area between 300 m and 900 m elevation. The up-country region is the area above 900 m elevation. Sri Lanka is generally described in terms of its terrain as comprising of a central highland mass surrounded by undulating to flat plans. It can be observed that about one third of the country is made up of hilly, mountainous or steep terrain.

Three major agro-climatic regions, wet, intermediate and dry zone, are further sub divided into 24 agro-ecological regions. According to the modern scheme of climatic zonation the wet zone of Sri Lanka falls within the humid tropics and the dry zone within the seasonally dry tropics. A small section in the northwest and southeast of the island falls within the semi-arid tropics. Four basic types of rainfall, namely, monsoon, convectional, cyclonic and orographic, are identified in the country (Panabokke, 1996). There are two main cropping seasons namely "maha" (northeast monsoon) - wet season from September to February and "yala" (southwest monsoon) - dry season from March to August. Due to availability of water, the "maha" is considered as the major cropping season.

The main annual rainfall in Sri Lanka varies from 2 500 to over 5 000 mm in the southwest of the island, while in the northwest and southeast an annual rainfall is less than 1 250 mm. Rainfall over most parts of the country follows a bimodal, seasonal pattern for the year. The minimum and maximum air temperature varies from 10 to 20°C in the Up-country Wet Zone (UCWZ) from 25 to 32°C in Low Country Dry Zone (LCDZ). The regional differences in temperature are mainly due to altitude with the temperature falling about 1°C for every 150 m rise in elevation.

2. Soils

The major soil types in Sri Lanka are Lateritic soils, the Red Earths, and the Alluviums. They cover 90 percent of Sri Lanka. The rest consisting saline soils and peat, are limited occurrence. Thirteen great soil groups have been identified in Sri Lanka namely Rhodustals, Tropaqualls, Haplustalf, Haplustox, Tropaquents, Natraqualls, Ustipsamments, Quartzipsamments, Pellusterts, Eutropepts, Rhodudults, Tropudults and Tropohemists (Panabokke, 1996).

3. Food production

3.1 Rice

Rice has been cultivated in Sri Lanka for well over 2 000 years. Sri Lanka has 0.734 million ha of rice lands (AgStat, 2004) of which 0.46 million ha (55 percent) are in the Low Country Dry Zone (LCDZ), 0.13 million ha (20 percent) in the low country intermediate zone (LCIZ), 9.14 million ha (25 percent) in the low country wet zone (LCWZ) and about 4.5 thousand ha are situated in up-country and midcountry areas. Rice cultivation accounts for a little more than 40 percent of the cultivated area. Presently, Sri Lanka produces 3.1 million tonnes of rough rice and national average yield of 3.8 t/ha⁻¹ (CBSL, 2004). Domestic annual rice requirement in the year 2000 was 3.11 million tonnes and it is estimated to be 3.46 million tonnes in the year 2010 (Abey Siriwardena and Sandanayake, 2000). Rice is grown in diverse soil and environmental conditions in throughout the country. Rice is grown in almost all the great soil groups and soil series found in Sri Lanka.

3.2 Vegetables

Approximately 60 000 ha of land is used for vegetable cultivation in Sri Lanka. (Maraikar *et al.*, 1996). Vegetable crops grown in Sri Lanka could be divided into two major groups, the local vegetables and the low country vegetables. Popular low country vegetables are okra (*Abelmoschus esculentus*), eggplant (*Solanum melongena*), luffa (*Luffa acutangula*), snake gourds (*Trichosanthes cucurbitina*), bitter gourds (*Momodica charantia*), cucumber (*Cucumis sativus*), pumpkin (*Cucurbita maxima*), and leafy vegetables, while, the exotic or up-country vegetables are bean (*Phaseolus vulgaris*), carrot (*Daucus carota*), beetroot (*Beta vulgaris*), cabbage (*Brassica oleracea*), leek (*Allium ampeloprasum*), and radish (*Raphanus sativus*). Low country vegetables are grown throughout the country with the exception of hill country areas, while up-country or exotic vegetables are mainly grown in the Up-country Wet Zone (UCWZ) and UCIZ. In general, vegetable cultivation is intensive and highly commercialized when compared to rice and other field crops (OFC) in Sri Lanka.

3.3 Root and tuber crops

Important root and tuber crops are potato (*Solanum tuberosum*), sweet potato (*Ipomoea batatas*) and cassava (*Manioc esculentus*). In general, potato is mainly grown in the UCIZ and UCWZ. Sweet potato, cassava and other root and tuber crops are grown in many parts of the country with the exception of the UCWZ. The extent of root and tuber crops grown in Sri Lanka is over 50 000 ha (AgStat, 2004).

3.4 Other Field Crops (OFC)

OFC, which include annuals of coarse grains, grain legumes, condiments and oil crops are grown in the dry zone. In general, the majority of OFC is cultivated under rainfed conditions. The extent of OFC grown in Sri Lanka is over 115 000 ha (AgStat, 2004).

4. Current status of fertilizer use

Significant use of high analysis chemical fertilizers for food crops in Sri Lanka started in early 1950, but widespread use of chemical fertilizer for food crops commenced in the year 1960. In order to promote balanced use of plant nutrients, fertilizer mixtures containing N, P and K nutrients had been recommended during early stages. However, the Department of Agriculture (DOA) changed its policy in 1990 by recommending straight fertilizers instead of fertilizer mixtures for food crops grown in Sri Lanka. This policy decision was taken in order to provide an opportunity for farmers to make

necessary adjustments in fertilizer application for optimum yield. However, the use of fertilizer mixtures containing N, P and K is more popular than straight fertilizers. This may be due to convenience in fertilizer use and lack of available straight fertilizers in the local market. The cost of fertilizer has increased tremendously over the last few decades and is still increasing. This situation has created financial difficulties particularly for small farmers. Fertilizer consumption by different crop sectors is presented in Table 1.

Table 1. Fertilizer consumption by crop sectors (mt)

Crop	1998	1999	2000	2001	2002
Paddy	251 880	321 032	262 362	284 488	356 174
Tea	182 339	164 139	200 254	182 033	185 059
Rubber	15 684	9 936	13 801	9 129	6 924
Coconut	37 667	39 161	33 942	29 531	38 452
OFC	44 182	42 617	40 467	34 320	42 035
EAC*	11 317	9 637	8 704	8 711	8 321
Tobacco	3 768	3 666	3 204	2 525	2 577
Others	14 515	26 147	29 797	28 770	30 675
Total	561 352	616 335	529 531	579 507	670 217
* EAC – refers to spices and beverage crops. Source: NFS, 2002.					

5. Types of plant nutrient used

5.1 Nitrogen fertilizers

At present, Urea is the most dominant form of nitrogen fertilizer used in Sri Lanka. It accounts almost 59 percent of the total fertilizers used in the country (NFS, 2002). In 1990, the DOA changed its policy of recommending sulphate of ammonia and switched to Urea fertilizers. Annually, almost 350 000 mt of Urea are imported and used either as a straight fertilizer or in mixed form. The introduction of heavy subsidy scheme for Urea by the government caused the tremendous increase in the demand for the said nitrogen fertilizer.

Sulphate of ammonia is considerably the second major nitrogen fertilizer used. However, sulphate of ammonia is only presently used for selected crops, and for the preparation of fertilizer mixtures found in the market. Annual imports of different fertilizers and their percentage share are given in Table 2.

Table 2. Annual fertilizer imports and their percentage share (mt)

Fertilizer	2001		2002	
	Quantity	Share	Quantity	Share
Sulphate of ammonia	71 214	16.04	75 136	13.24
Urea	258 785	58.27	337 906	59.48
Imported rock phosphate	12 250	2.76	11 778	2.07
Triplesuperphosphate	14 557	3.28	40 495	7.13
Muriate of potash	74 737	16.83	95 156	16.75
NPK-5:15:15	6 600	1.49	—	—
Kieserite	4 411	0.93	5 402	0.95
Sul-Po-Mag	800	0.18	200	0.04
Commercial epsom salt	405	0.00	538	0.09
Zinc sulphate	305	0.07	599	0.11
Sulphate of potash	195	0.04	214	0.04
Diammonium phosphate	154	0.03	648	0.11
Total	444 112	100.00	568 072	100.00
Source: NFS, 2002.				

5.2 Potassium fertilizers

Muriate of Potash (MOP) could be the second most dominant fertilizer used in the country. The annual importation is estimated to be at the level of over 95 000 mt. As a whole, paddy field sector uses almost 36 percent of imported MOP, while other field crops use about 8 percent and vegetable and other crops use about 37 percent. (NFS, 2002). Another minor source of potash, Sulphate of Potash (SOP) is used in limited scale for chloride sensitive crops.

5.3 Phosphorus fertilizers

Triplesuperphosphate (TSP) is the most dominant phosphorus fertilizer used for short-duration crops such as rice, vegetables and other crops. Almost 41 000 tonnes of TSP are imported into the country annually, 80 percent of which are utilized for paddy cultivation. The balance is used mainly for vegetables, OFC and fruit crops, except in wet zone areas of Sri Lanka. Phosphorus is the most limiting plant nutrient in Sri Lankan soils. Notably, application of TSP has shown beneficial effects for

crop growth and for subsequent high yields (Wijewardena, 1994). Local rock phosphate named eppawala rock phosphate is mainly being used for fruit crops grown in the wet zone of Sri Lanka. However, a number of research studies conducted in relation to eppawala rock phosphate revealed that it cannot match the P-supplying ability of TSP for annual crops, and its direct application as a source of P fertilizer for food crops will not be beneficial (Wijewardena, and Amarasiri, 1990; Wijewardena, 1998).

5.4 Compound fertilizers

The most popular compound fertilizers used in Sri Lanka are granular NPK with trace elements. One of the most important compound fertilizers used was NPK of the formulation 5:15:15 supplied under the Japanese Food Production Grant for paddy cultivation particularly in the wet zone of the country. This formulation of compound fertilizer was very popular among paddy farmers especially in the Low Country Wet Zone (LCWZ) where annual rainfall is over 2 500 mm. Various types of compound fertilizers with different formulations are also available in the local market despite the absence of large imports of NPK compound fertilizer.

5.5 Magnesium fertilizers

Most popular magnesium fertilizer is dolomite which is available locally as calcium magnesium limestone. Dolomite is often used as a soil ameliorant for acid soils. It is mainly used for potato, vegetables and other horticultural crops. However, kieserite is also imported in limited quantities mainly for horticultural crops. Similarly, epsom salt is used to correct widespread magnesium deficiencies in fruits, vegetables and OFC.

5.6 Zinc sulphate

The Department of Agriculture in 2001 recommended application of zinc sulphate for rice grown in dry and intermediate zones. In addition, zinc deficiency is a common problem in fruit cultivation particularly of citrus. As a result, almost 600 mt of zinc sulphate was imported in 2002 to Sri Lanka.

5.7 Special fertilizers

Special fertilizers used in Sri Lanka are NPK formulations with trace elements, micronutrient, mixtures for hydroponics, fertigation mixtures and fertilizer with growth promoting substances (Table 3). These fertilizers are used for vegetables, floriculture crops and other horticultural crops. Most of these fertilizers are comparatively high value fertilizers imported in small quantities.

Table 3. Imports of special fertilizer in the year 2002

Fertilizer	Liquid (liters)	Quantity granules (mt)	Powder (mt)
NPK	2 296	179.5	21.0
NPK + TE	8 910	636.5	7.9
TE	6 001	47.1	18.9
Fertigation mixture	—	—	11.5
Hydro phonic mixture	—	—	12.0
TE = Trace Element Source: NFS, 200			

5.8 Liquid fertilizers

There are more than hundreds of types of liquid fertilizer that are available in the market. They are either organic fertilizers or fertilizers containing both organic and chemical fertilizer. Even though the DOA has not recommended liquid fertilizers, some farmers are using them for crops such as vegetables, onion, potatoes, floricultural crops, etc. By application of foliar fertilizers, farmers expect to obtain attractive product, which can fetch high prices in the market rather than get their yield increased.

6. Trends in plant nutrient use

Some economic factors play an important role in determining the quantity of chemical fertilizers used by farmers vis-à-vis yield response to added fertilizer, farm gate price of produce, price of fertilizer, etc. Prior to 1966, use of low analysis fertilizer was a common feature for rice. Rock phosphate or bone meal and sulphate of ammonia were the sources of phosphorus and nitrogen, respectively. However, during the period of 1967 to 1978 Urea became the major source of nitrogen. Similarly, TSP replaced rock phosphate. The period from 1979 to 1989 indicates a high rate of growth in fertilizer use for paddy. Introduction of high fertilizer responsive varieties and the enhanced area under irrigation are some factors that contributed to this growth. A setback in 1996 was observed due to the price shock after the removal of fertilizer subsidy. It took almost 3-4 years to recover from this setback. Use of sulphate of ammonia was recommenced during the period 1990-1994 due to transport restrictions on Urea to north and to the east because of security reasons. However, in 1990 the DOA changed its policy of recommending fertilizer mixtures for food crops and switched to straight fertilizers due to increased flexibility in terms of adjusting amounts of nutrients to suit the crop and soil. Still, a large number of fertilizer mixtures continue to be marketed in Sri Lanka. Most farmers prefer the mixtures, probably because they are less cumbersome to use, although they can be more costly than straight fertilizers per unit of nutrient. In general, for the past several years, the use Urea fertilizer maintained an increasing trend.

6.1 Rice

Paddy consumes the largest part of chemical fertilizers. It accounts for approximately 50 percent of the overall use of chemicals fertilizers in Sri Lanka (Table 1). The main fertilizers used for paddy cultivation are Urea, TSP and MOP. Though many farmers in dry and intermediate zones use recommended level of chemical fertilizers, farmers in wet zone use chemical fertilizers less than the recommended level. These facts indicate that overall fertilizer use is below the recommended level. This may be due to using less or no basal fertilizer for paddy cultivation by the farmers growing paddy in marginal lands and in cultivation under rainfed conditions. The cost of cultivation of paddy in Sri Lanka is increasing from season to season. As a result, use of balanced NPK for rice has been decreased. In addition to NPK fertilizers, the DOA has recommended once-a-year application of zinc sulphate at the rate of 5 kg ZnSO_4/ha for rice cultivation. Popularization of straight fertilizer and promotion of Integrated Plant Nutrition System (IPNS) in paddy cultivation have been conducted by the DOA in recent past. It has been observed that there is a slow, but a steady growth in straight fertilizer use for paddy during the last few years. However, paddy fertilizer mixture such as 'V mixture (5:15:15) and TDM (30:0:20) were also in wide use by rice farmers. Paddy production depends on several factors and imbalances in fertilizer usage need to be investigated for future improvements, as there had been a set back in the NPK ratios in the fertilizer application of paddy. In this regard, production increase programme conducted by the DOA namely "Yaya" demonstrations programme, had contributed positively to enhance higher productivity in paddy cultivation.

6.2 Vegetables

Vegetables have become an important component in almost every cropping system. Of the reasons for widespread cultivation of vegetables, the most important is that net returns from vegetable production are higher than returns from rice and most other field crops. Unlike the general pattern of low fertilizer used in paddy and OFC farmers, most vegetable farmers use high quantities of chemical as well as organic fertilizers. The level of fertilizer applied by the farmers to the vegetables crops is almost two to three times the quantity recommended by the DOA (Wijewardena, 2001). This may be due to favourable crop price relationship for vegetable crops. As a result, net returns from vegetable cultivation are much higher than in rice and OFC, making chemical fertilizer a relatively inexpensive input. Various types of liquid fertilizers are also used regularly as foliar fertilizers.

6.3 Other field crops

Since OFC are mainly grown under rainfed conditions, areas under these crops may vary season to season. The area under OFC covered approximately 76 000 ha in rainy season – Maha (northeast monsoon) followed by 30 000 ha in dry season – Yala (southwest monsoon). In general, fertilizer use for OFC crops could be considered as little or no fertilizer application at all. It indicates that OFC

crops mainly depend on soil nutrients rather than on fertilizers. However, for cash crops such as onion and chili, high rates of fertilizers are used. Overuse of chemical fertilizers for onion has been reported in some areas, such as in Kalpitiya.

6.4 Fruit crops

Nutrient management can be considered as one of the most neglected aspects for the majority of fruit crops grown in Sri Lanka. In addition, nutrient management recommendations for the majority of fruits are not based on proper research conducted in the country. Hence, due attention should be given to nutrient management practices in relation to fruit crops. Except few commercial growers, most small-scale farmers rarely use chemical fertilizers for many fruit crops. In general, fertilizer prices currently prevailing in Sri Lanka are not affordable to marginal fruit growers. Even most commercial growers use various types of fertilizer mixtures available in the market such as coconut, paddy and vegetable fertilizer mixtures. However, after establishment of the Horticultural Crop Research and Development Institute in Sri Lanka, research studies have been initiated to solve such problems in the fruit crop sector.

7. Future trends

Price factor of fertilizer is major constraint in promoting chemical fertilizers in Sri Lanka, particularly for food crops such as rice and OFC. As a result, production and use of local fertilizers is a strong future possibility. In view of high price of triple superphosphate there is a necessity for utilization of local eppawala rock phosphate (ERP) for food crops. In this regard, steps have been taken to produce single superphosphate (SSP) locally using ERP.

8. Types of fertilizer produced locally

Sri Lanka depends on imports of chemical fertilizer to meet the requirement. In the year 2002, almost 92 percent of the total supply of chemical fertilizer, except for eppawala rock phosphate (ERP) and Dolomite, was made available through imports. Rock phosphate deposit at eppawala was discovered in 1971. Its tonnage has been estimated as 40-60 million mt of apatite rock. There are two forms of rock phosphate fertilizers being produced in Sri Lanka, namely ERP and high grade eppawala rock phosphate (HERP). HERP, could be considered as an improvement, which is processed by crushing pure apatite only. At present, annual production of ERP is 41 000 mt (Table 4), while HERP is almost 6 000 mt (Lanka Phosphate Ltd., 2005 Unpubl.). Though Sri Lanka imported high citric acid soluble rock phosphate, a few years ago after introduction of HERP, substitution of imported rock phosphate (IRP) with HERP was done. As a result, rock phosphate fertilizers are not presently being imported to the country. With the increased demand for plantation crops and also with enhanced promotional efforts, the local supply of ERP had reached the highest level achieved so far. Eppawala rock phosphate is used for perennial crops, such as tea, rubber, coconut, fruit crops etc.

Table 4. Annual production and issues of local rock phosphate (mt)

Operation	2000	2001	2002	2003
Production	35 085	35 769	37 664	41 351
Issues	35 805	35 441	38 412	40 170

Experiments have been conducted during the past three decades to find out the suitability of ERP for annual crops (Nagarajah *et al.*, 1979; Wijewardena and Amarasiri, 1990; Wijewardena, 1998). Removal of phosphate by annual crops like potato and vegetables are high as they generate a high biomass within a short period of time. In addition, the shallow root system found in these plants requires soluble types of P fertilizers. As a result, direct application of rock phosphate sources are not suitable for annual crops due to their low solubility in soil solution. However, partial acidulated rock Phosphate (Wijewardena and Yapa, 1998), mixtures TSP and rock phosphate combinations, combined application of rock phosphate and organic manure showed beneficial effects for all annual crops including rice, potato and vegetables. The Lanka Phosphate Limited (LPL) has produced limited quantity of single superphosphate (SSP) from ERP, which contain more soluble P than ERP. At present, experiments are in progress to study the suitability of SSP as a source of P for annual crops such as rice, vegetable and OFC.

Dolomite is mined and crushed by private firms in the Central Province of Sri Lanka. The estimated use of dolomite as a fertilizer or soil ameliorant has arisen to about 20 000 mt. It is regularly used as

a liming material for tea, fruit crops, vegetables and potatoes.

9. Cost of fertilizer products

Urea is subsidized in Sri Lanka. However, the local price of all other fertilizers including TSP, MOP and sulphate of ammonia was determined by the open market forces. The competition among the number of wholesalers also assisted in keeping the open market price competitive. The demand for Urea, owing to its subsidized price, was maintained at a reasonably high level. The price of other fertilizers increased in the local market due to increase of oil prices in the world market. In general, however, price of Urea in Sri Lanka continues to be increased. The retail price of major fertilizer in selected ESCAP countries during the year 2002 indicated that comparative price of fertilizer appear to be high in Sri Lanka (NFS, 2002). Retail prices of fertilizers are presented in the Table 5. This table indicates that among commonly used fertilizers in Sri Lanka, TSP is the most expensive in the local market followed by MOP and Urea.

Table 5. Average retail prices of important fertilizers (US\$/mt)

Fertilizer	6 th July 2000	12 th Feb. 2001	18 th July 2001	15 th July 2002	10 th Oct. 2002
Urea	70	70	70	70	110
Sulphate of ammonia	106	106	136	136	156
Triplesuperphosphate	192	192	172	210	210
Muriate of potash	165	165	186	199	199
Local rock phosphate	41	41	45	36	56

Source: NFS, 2002.

10. Fertilizer subsidies

The subsidy scheme for fertilizer was initiated more than a decade ago. However, since 1997, only Urea was subsidized in Sri Lanka. At present, subsidy for Urea is approximately US\$230 per mt. As a result, an increasing trend in use of nitrogen fertilizer was observed. In addition, a gradual and substantial reduction in use of P and K fertilizer is evident. This implies a deterioration of NPK ratios in recent years. The international market price of Urea also increased during the year 2004. Application of the US dollar against the local currency, removal of national security levy on fertilizer and introduction of value added tax (10 percent) contributed to increase in Rupee expenditure on Urea. In keeping with the fluctuation in the international market price of Urea, reasonable import prices and the corresponding rates were fixed for implementation of the subsidy. At present, there is a substantial increase in the quantity imported under the subsidy scheme due to enhanced demand for fertilizer. However, fertilizer subsidy is a very strong political issue in Sri Lanka.

Recently elected government in Sri Lanka has decided to revise the subsidy scheme and implement a fixed subsidy for all three fertilizers such as Urea, TSP and MOP. Under this proposal, a 50 kg bag of any fertilizer will be US\$3.5. This new scheme will be implemented with effect from 5 December 2005. According to this new scheme, subsidy will be given only for straight fertilizers. Estimated cost for proposed subsidy will be US\$0.12 billion per year. However, the new proposal is targeted to small farmers in the country.

11. Depletion of soil organic matter

Soils in Sri Lanka are low in organic matter (Panabokke and Nagarajah, 1964; Wijewardena, 1995; Wickramasinghe and Wijewardena, 2003). Long-term experiments conducted over eight (8) seasons with rice (J.D.H. Wijewardena, 2001 Unpubl.), potato and vegetables (Wijewardena, 1993;

Wijewardena and Yapa, 1999) showed that there was no increase of organic matter in soil by application of organic manure 4 and 10 t/ha per crop, respectively. Research shows that the majority of soils in main rice growing areas are low in organic matter (Table 6). As a result, many soils found in Sri Lanka are low in CEC. Cation exchange capacity levels in majority soils are lower than 10 cmol/kg (Panabokke, 1966). Under such conditions, retention of plant nutrients is low. In general, organic manures are useful in conserving soil fertility. As a result, seasonal application of available organic manure sources showed beneficial effects of crop growth as well as overall soil fertility (Wijewardena 1993; 2000). Hence, a policy decision was taken by the Ministry of Agriculture and the Department of Agriculture in 1999 to promote IPNS technology among farmers in Sri Lanka. This technology showed economical benefits as well as soil fertility improvement in relation to various food crops grown in Sri Lanka (Wijewardena and Yapa 1999; Dissanayake, 2000).

National average yields of paddy and several other food crops have been stagnating over the last decade (Wickramasinghe and Wijewardena, 2003) requiring a new approach on how to increase at the national yield levels. Reasons for such problem may be due to declining of organic matter content, imbalance of major plant nutrients, and inadequacy of various micronutrients in most of the cultivated lands in Sri Lanka. As a result, these factors have adversely affected the overall soil fertility in cultivated lands in Sri Lanka. Soil organic matter content is one of the key parameters which influence the soil fertility and productivity. However, it should be noted that increase of organic matter in soils is a difficult task due to prevailing high temperature in many parts of the country. In this regard, the seasonal applications of organic materials are necessary. Rice straw, animal manure, green manure, city wastes, crop wastes, etc. could be used as sources of organic manure to maintain the organic matter content in soil. There was an indication that in all the zones, except in the low country wet zone (LCWZ), highest rice production occurs on soils with an organic matter content of around 2 percent. However, at the LCWZ, soils with an organic matter content of 4-6 percent gave better crop response (Dissanayake *et al.*, 1993). In addition, high content of organic matter was recorded in wet season (Maha) than in dry season (Rezania *et al.*, 1992). However, a slight increase in soil organic matter content was evident with more intensive cropping of potato and vegetable cultivation in the up-country areas of Sri Lanka where low temperatures are prevailing due to high elevation.

Table 6. Soil organic matter content (percent) in major rice growing districts

District	1964*	1984**	2000***	
	Organic matter content (%)		Organic matter content (%)	
			<1%	1-3%
			No. of sites	
Ampara	0.69-1.72	1.4	45	25
Anuradhapura	0.86-2.58	1.4	3	87
Polonnaruwa	0.86-2.58	1.9	20	75
Kurunegala	0.68-1.72	1.9	89	9
Hambantota	0.72-3.87	2.5	34	49
Badulla	2.59-6.45	2.0	26	40

* Panabokke, 1964; ** Rezania, 1984; *** Soil Testing Programme of the DOA, 2000.
(Adopted from: Wickramasinghe and Wijewardena, 2003)

12. Decline of plant nutrients

Agricultural lands diminish in crop production potential or suitability for crop production through various types of land degradation in Sri Lanka (Nayakekorale, 1998). Being an agricultural country, Sri Lanka has to place much attention on soil degradation problem in consideration of the low per capita land availability at present. The directly available land for human use was estimated at 0.15 ha/person (Madduma Bandara, 2000). Soil fertility decline in Sri Lanka is mainly due to depletion of soil organic matter as well as loss of plant nutrients etc. Soil analytical studies conducted in various parts of the country revealed that low plant nutrient content is a major threat to crop production in Sri Lanka (Wijewardena, 1995; Nayakekorale and Prasantha, 1996). The depletion of soil nutrient due to leaching and run off could be considered as major course of fertility decline. Thus, many agricultural farming systems are becoming non-profitable to farmers.

Rice growing soils in Sri Lanka have very low content of P in available form (Tables 7 and 8). In general, P-fixing capacity of majority of soils is also high. Phosphorus deficiency in rice soils of the midcountry has been previously reported by Panabokke and Nagarajah (1964) and Maraikar *et al.*, (1983). Similarly, low available P in rice soils of the LCWZ (Wijewardena, *et al.*, 1998) has been reported. Critical level of available soil P in the soils of the LCDZ and LCIZ is about 3-5 mg/kg, while it is around 10-15 mg/kg in soils of midcountry wet zone (MCWZ) and MCWZ, to achieve a yield response to added P. Rice farmers apply low quantities of P due to high cost of TPS in Sri Lanka. Soil analysis conducted in different areas show that 63-80 percent soils are low available P (Olsen P) except in some soils which grow potato and vegetables in rotation with rice (Wijewardena, 1999) in the up-country intermediate zone (UCIZ).

Table 7. Uncultivated and cultivated soil characteristics in the up-country

Characteristic	Uncultivated soil	Cultivated soil
pH (1:2.5; Soil: H ₂ O)	4.0-5.5	4.5-6.0
OM (%)	1-2	2-4
Total N (%)	0.1-0.15	0.2-0.3
Olsen P (mg/kg)	2-10	20-400
Exchangeable K (meq/100 g soil)	0.1-0.2	0.2-0.5
Source: Wijewardena, 1995.		

Table 8. Range of available P in rice, vegetable and fruit growing soils of the LCWZ

Rice*		Vegetables**		Fruits***	
Soil P (mg/kg)	No of sites (%)	Soil P (mg/kg)	No of sites (%)	Soil P (mg/kg)	No of sites (%)

<5	33	<10	43	<5	41
5-10	36	10-20	30	5-10	28
10-20	26	20-30	3	10-20	17
>20	5	>30	24	>20	14

Source: * Wijewardena *et al.*, 1998; ** Wijewardena *et al.*, 1999a; *** Wijewardena *et al.*, 1999b.

Similarly, low available P in upland soils is also common in Sri Lanka: in soils of the UCWZ (Kumaragamage *et al.*, 1999), UCIZ (Wijewardena, 1995; Wijewardena, 1996), and UCWZ (Wijewardena *et al.*, 1999). High P-fixing capacity in up-country vegetable growing soils has been reported by Withana and Kumaragamage (1993). A study conducted with 70 soil samples by Wijewardena *et al.* (1999) reported that majority of soils collected from vegetable growing soils in the LCWZ had available P below 20 mg/kg (73 percent sites), which is considered as the low content of P according to the criteria set by the DOA in 1995. Similarly, Kumaragamage *et al.* (1999) also reported the low P availability in soils of the LCWZ.

However, build up of P could be seen in many intensive vegetable growing soils in Sri Lanka. This condition is caused by indiscriminate use of organic and chemical fertilizers. Build up of phosphorus in some potato and exotic vegetable growing soils in the up-country area particularly in UCWZ and UCIZ has been reported (Jeewanathan *et al.*, 1995; Wijewardena, *et al.*, 1996; Wijewardena, 1999). Similarly, occurrence of high available P due to use of high rates of deep litter as well as chemical fertilizers was observed in leafy vegetable growing soils in the LCWZ. Soil analysis revealed that available P (Olsen's) in these soils is as high as 2 500 mg/kg (J.D.H. Wijewardena, 2004 Unpubl.).

Field crops in Sri Lanka are traditionally cultivated in the dry and semi dry areas. Field crops on rainfed uplands receive inadequate fertilizer phosphorus and as a result, soils are relatively low in available P. However, crops such as chilli and onion on irrigated uplands receive excessive fertilizer P, thus, soils become generally high in available P.

In Sri Lanka, most fruit crops are grown in home gardens under rainfed conditions. As a result, use of fertilizer for fruit crops is limited. Hence, fruit growing soils in Sri Lanka are low in phosphorus (Wijewardena *et al.*, 1996; J.D.H. Wijewardena, 2003 Unpubl.).

All rice growing soils have different K supplying capacity. According to Panabokke and Nagarajah (1964), K status in rice growing soil was in a range of 39-59 mg/kg. The exchangeable K content of rice growing soils in the LCWZ ranges between 27-94 mg/kg. In addition, coarse textured soils found in the LCDZ also contain the lower exchangeable K content at 47 mg/kg. In some cases, K in these soils has been reported to be as low as 12-24 mg/kg (Weerasinghe, 1991). Almost 50 percent of the rice growing soils in the LCWZ have exchangeable K content of less than 58 mg/kg, which is considered as the critical soil K level for mineral or alluvial soils (Rezania, 1992). It has been also observed that there is a declining trend in these soils when rice is grown continuously. According to Wijewardena, *et al.*, (1998), 48 percent of soils in the LCWZ contain exchangeable K below 78 mg/kg of soil, while remaining soils contain more than 78 mg/kg of exchangeable K.

In the UCIZ, a total of 150 soil samples collected in 57 fields revealed exchangeable K values exceeding 300 mg/kg in soils fertilized for potato and vegetable crops for few decades (Wijewardena, 1999). Since P is also high in these soils, the DOA has recommended only N for rice grown in this region. Similarly, high K contents in rice-vegetable cropping systems were also reported in the MCIZ (Joseph *et al.*, 1988).

A study conducted by Wijewardena *et al.*, (1999) reported low exchangeable K even in vegetable and tuber crops growing soils in the LCWZ. They showed that based on soil K standards introduced by the DOA (1997), 83 percent sites had a low soil K. Hence, application of K fertilizers is necessary to improve soil fertility in the LCWZ vegetable and tuber crops growing soils to obtain high crop yields in the LCWZ. Similarly, low exchangeable K in soils of the LCWZ was also reported by

Kumaragamage *et al.*, (1999). Soils found in LCDZ and LCIZ are containing reasonable amount of exchangeable K. However it should be noted that K in LCDZ are higher than in LCIZ (Wijewardena, 2002).

Soils found in Kalpitiya peninsula were used to cultivate vegetable, chili etc. Though soils are poor in soil fertility, chemical fertilizers are used extensively for crops grown in this region. As a result, build up of some plant nutrient could be seen in these soils. However, exchangeable K content in these soils is low. This could be expected due to the sandy nature of soils found in this area.

Investigations conducted (Joseph *et al.*, 1988) in midcountry area revealed that accumulation of K could be seen in cultivated soils. Farmers use high rates of both organic and chemical fertilizers for vegetables and chemical fertilizers for vegetable crops grown in this region.

In both UCWZ and UCIZ, vegetable and tuber crops cultivation is intensive and highly commercialized. In the UCWZ, vegetables are cultivated throughout the year in rotation with potato. In the UCIZ, vegetables are grown under upland rainfed, irrigated and lowland rice-based cropping systems. Due to the hilly nature and high rainfall in this region, soils are rather poor in plant nutrient contents. As a result, inherent soil fertility in Ultisols (Kumaragamage *et al.*, 1999) is very low. However, Ultisols in the up-country contain reasonable quantity of K in exchangeable form (Wijewardena 1995a). In addition, most farmers in the up-country utilize inorganic fertilizer mixtures in combination with large amount of organic manures with particular emphasis on the use of poultry litter and cattle manure. Wijewardena (1995) compared the characteristic of uncultivated and cultivated soils in the up-country of Sri Lanka (Table 7). A total of 290 soil samples analysed by Wijewardena *et al.*, (1996) reported that 44 percent of soil samples had K up to 160 mg/kg, 50 percent of soil samples had K between 160-400 mg/kg and only 6 percent soil samples had K more than 400 mg/kg. A sampling programme conducted after rice crop included 68 fields for a total of 161 soil samples taken at depths of 0-15 and 15-30 cm respectively in the rice based cropping system in the UCIZ showed the available K in excess of exchangeable K of 156 mg/kg in 71 percent of the fields and relatively high K content even in the subsoil of 15-30 cm depth (Wijewardena, 1999).

13. Micronutrients

Rice growing soils particularly in dry and intermediate zones are deficient in Zn and Cu (Bandara and Silva, 2000a; 2000b). In addition, poorly drained rice growing soils of midcountry wet zone of Sri Lanka are also deficient in zinc (Nagarajah, *et al.*, 1983). Significant response to Zn addition was observed in dry and intermediate zones. Zinc application improved the grain yield of rice from 10-15 percent (Bandara and Silva, 2001). Hence, Zn was recommended by the DOA for rice cultivation at the rate of 2.5 kg Zn/ha as ZnSO₄ · 7H₂O. Study conducted by Deb *et al.* (1993) reported that majority of rice growing soils are also deficient in micronutrients such as Zn, Mn, Cu. Similarly, they reported toxicity levels of Fe in paddy soils in the LCWZ. In general, 30 000 ha of cultivated rice lands in the LCWZ considered as having a potentially iron toxic condition (Bandara and Gunathilaka, 1997). This problem may be associated with excess soluble iron, low pH, low soil fertility, poor drainage etc. (Ponnamperuma, 1958). Some chemical characteristics of iron toxic rice growing soils are shown in Table 9. Application of balance chemical fertilizers, particularly K fertilizer in combination with organic manures, is a very important soil management practice. Application of liming materials such as dolomite, lime and deep litter has shown effective measures of minimizing iron toxicity in rice cultivation (J.D.H. Wijewardena, 2004 Unpubl.).

Table 9. Some chemical properties of rice growing soils in the LCWZ

Property	Range	Average	SD
pH (1:2.5; Soil: H ₂ O)	4.2-5.1	4.5	0.24
EC (dS/m)	0.019-0.042	0.028	0.0067
OM (%)	2.8-7.9	4.9	1.3

Exchangeable K (mg/kg)	15.5-65.7	31.2	12.4
Olsen P (mg/kg)	4.8-18.3	9.7	3.54
NH ₄ OAc Exchangeable Fe (mg/kg)	377-923	626	130
Source: J.D.H. Wijewardena, S.P. Gunaratna and S. Weerasinghe, 2004 Unpubl.			

Studies on micronutrients are rather limited in Sri Lanka. However, an investigation conducted by Wijewardena and Kannangara (2003) reported that majority of vegetable growing soils of LCWZ contain adequate amounts of Zn (>0.6 mg/kg), Mn (3.5 mg/kg) and Fe (2.5 mg), but soils were deficient in Cu (<0.2 mg/kg).

In addition, citrus growing soils in the Uva region are deficient in available Zn, Fe and Cu contents (Wijewardena, 2004). As a result, application of Zn, Fe and Cu fertilizers and organic manures could be considered as important crop and soil fertility management practices to obtain high yields of citrus growing soils in Sri Lanka. Zinc and boron deficiency in citrus and papaya, respectively, is a widespread occurrence in fruit crops grown in Sri Lanka. A soil analytical programme conducted with 120 soil samples collected from major citrus growing region of Sri Lanka revealed that citrus growing soils are deficient in micronutrients such as Zn, Cu, Fe, and Mn. (Wijewardena, 2004). Similarly, fruit growing soils in the LCWZ are deficient in Cu (Wijewardena *et al.*, 2001).

14. Organic fertilizer use in Sri Lanka

Since ancient times, manuring has been considered as one of the most important techniques to increase and maintain soil fertility. Soil organic matter plays an important role in soil fertility. Rice straw could be considered as a very valuable organic fertilizer material for rice grown in Sri Lanka. Since, many rice soils in Sri Lanka are low in soil fertility it should be noted that straw can be successfully used to improve the long-term soil fertility in many parts of the country. However, a large number of farmers burn the straw at the threshing site especially when there is no alternative way of disposal. Majority of Sri Lankan farmers are fully unaware of the value of rice straw as a fertilizer material and a large quantity of rice straw is wasted. Despite the recommendation from DOA, there is still low acceptance by farmers to apply rice straw as a source of organic manure. Some of the reasons for this are the following: cost and inconvenience of transporting the bulky material to the field, unavailability of simple and inexpensive methods to apply straw to the fields in the manner which does not hinder land preparation without causing adverse effects on the rice crop.

In general, rice straw is a valuable source of carbon, potassium, silicon, etc. It is also a convenient source of organic manure in Sri Lanka, aside from being considerably the cheapest. Rice straw thereby could be considered as a very important organic fertilizer material for rice grown in Sri Lanka (Dissanayake 2000). Long-term field experiments conducted at the LCWZ revealed that straw application in combination with NPK can increase the yield of rice as well as over all soil fertility (J.D.H. Wijewardena, 2000 Unpubl.). Hence, rice straw should be applied in combination with recommended chemical fertilizers. In order to get maximum benefits, the straw must be incorporated into the soil by 2-3 weeks before the land preparation or 3-4 weeks before planting of rice.

On the other hand, the use of animal wastes for rice cultivation is very limited. Unlike for cash crops, the costs of transportation of animal manure are too high to permit their application in rice cultivation. These considerations suggest that organic manure should be generated in the field itself or in its near vicinity if the practice is to be accepted by farmers. In addition, sources of animal manure are expensive due to high demand by vegetable and potato growers in the country. As a result, rice farmers are not in a position to purchase organic fertilizers such as deep litter, cattle manure, goat dung.

Alternatively, green manures have been added to rice fields for many years. Field experiments have shown that chemical fertilizer can be reduced by addition of green manure. However, use of green manure in rice cultivation is not widespread primarily due to limited availability of suitable plant materials. Often it is applied to nursery area or to parts boarding the high lands. Experiments with *Gliricidia maculata* and *Tithonia diversifolia* showed that 9 t/ha of these green manure in combination with 59 kg/ha NPK fertilizer gave rice yields more than the recommended NPK levels (Nagarajah, 1999). In addition, 4 seasons long-term experiment with *Tithonia diversifolia*, showed that the

amount of NPK fertilizer added to rice can be substantially reduced. Studies on the use of *Sesbania sesban* and *S. rostrata* as *in situ* green manures for lowland rice showed that both species can supply up to 50 percent of the recommended N fertilizer.

It is a common practice to apply organic sources, such as cattle manure and poultry manure, when growing vegetable and potato throughout the country. The up-country is considered as the region where main potato and vegetable growing area of the country. Potato and vegetable cultivation in the up-country is intensive and highly commercialized. A unique feature observed in the up-country of Sri Lanka in potato and vegetable cultivation is the use of animal manure such as cattle and poultry manure. Poultry manure, which has to be transported over a long distance, approximately 200-250 km from west and Northwestern part of the country, is commonly used as a fertilizer at the rate of 10-15 t/ha per crop (Wijewardena 1993; Wijewardena, 1995). Cattle manure is used as a fertilizer at the rate of 20-30 t/ha per crop (Maraikar *et al.*, 1996). In the up-country, it could be considered more as farmyard manure rather than pure cattle manure because it contains large quantities of grass, straw and leftovers of bedding materials. However, farmers in the up-country prefer to use farmyard manure than pure cattle manure. Cultivation of potato and vegetables has never included the application of cattle or poultry manure.

In vegetable cultivation, sunken beds are prepared and filed with cattle manure and covered with layer of soil. Then, farmers apply chemical fertilizer mixtures, mixed with soil and irrigated 1-2 days before planting. In the case of potato, cattle manure or poultry manure is spread along the furrows and are covered with a layer of soil. Here, too, chemical fertilizer mixtures are applied and irrigated 1-2 days before planting potato tubers.

It is very important to note that farmers in the up-country apply poultry manure 3-4 days prior to planting vegetable and crops and irrigate for 2-3 days if rains are not experienced within this period. By this practice, farmers avoid the harmful effects of poultry manure on their crops at the initial stage. In addition, farmers apply chemical fertilizer mixtures containing N, P and K or straight fertilizers 1-2 days prior to planting and mix with previously applied poultry manure. After irrigating for 1 or 2 days, poultry manure is mixed with chemical fertilizers before planting the crop. During the growing season, farmers top-dress their crops with NPK mixtures or TDM or Urea.

15. Use of foliar fertilizers

Potato and vegetable growers, particularly in the up-country, apply different kinds of foliar fertilizers during the growing period at 6-7 days intervals at a high rate of 12 l/year/ha (Rezania *et al.*, 1989). In general, foliar fertilizer is applied to vegetable crops grown in this region even just before harvesting. By these applications, farmers expect to obtain attractive vegetables, which can fetch high prices in the market rather than get their yield increased.

16. Use of liming materials

Application of liming materials such as dolomite and burnt lime is also a common practice for potato and vegetable growers particularly in the up-country areas. Despite the recommendation of DOA to apply such materials at 2 t/ha/year if soil pH is below 5, farmers still apply almost 650 kg/ha (Wijewardena, 2001).

17. Nutrient balance

Rice is a biannual crop and it removes considerable amounts of nutrients especially potassium and silica, which are found in straw and in grain. The total removal depends on the yield. Nutrient removed by rice crop indicates a negative balance of P and K even with the application of DOA recommended levels (Table 10).

Rice crop removes more P and K than quantities applied as chemical fertilizers. However, this situation could be improved by application of organic manures recommended by the DOA (Wickramasinghe and Wijewardena, 2003). It indicates that nutrient removed needs integrated use of organic and chemical fertilizers to compensate for the nutrient loss (Dissanayake, 2001).

Table 10. Nutrient balance in different rice growing environments of Sri Lanka

Area/Nutrient	Total removal (kg/ha)	Amount recommended (kg/ha)	Balance (kg/ha)

LCDZ				
	N	100	100	0
	P	18	13	-5
	K	106	28	-78
Up-country or Midcountry				
	N	100	60	-40
	P	18	13	-5
	K	106	28	-78
LCWZ				
	N	100	55	-45
	P	18	13	-5
	K	106	53	-78
In straw: N = 0.74 percent; P = 0.1 percent; K = 1.81 percent In grain: N = 1.26 percent; P = 0.26 percent; K = 0.32 percent				

18. Farm budgets in different cropping systems

The increasing total cost of production and declining profit have become significant issues particularly in rice and OFC. In high potential areas, nominal cost of production has doubled from US\$0.04/kg in 1987/88 maha season to US\$0.08/kg in 1998/99 maha season (Jayawardena, 2003). This analysis indicates that the real cost production per kg of paddy has become more prominent in low potential areas, attributing to decreased productivity. Similar trend could be seen in many OFC crops other than chili and onion. However, it is interesting to note that vegetables have become an important component in almost every cropping system. Of the several season for widespread cultivation of vegetables, the most important is that net returns from vegetable production are higher than returns from rice and most other field crops (Table 11).

Table 11. Comparison of net returns from cultivation of different crops per hectare (US\$)

Crop	Cost of cultivation	Value of product	Net return
Rice	45.7	75	29.3

Potato	424.3	825	400.7
Tomato	123.3	188	56.7
Bitter gourd	123.6	160	36.4
Bean	144.8	280	135.2
Green gram	213.5	230	16.5

19. Soil test-based fertilizer recommendations

The soil test-based fertilizer recommendation programme introduced by the DOA has contributed, to some extent, to prevent build up of P and K in the cultivated fields as well as to minimize groundwater pollution (Tables 12 and 13). Under this programme, any farmer can get his soil sample tested and obtain a site specific P and K fertilizer recommendation for rice, vegetables, tuber crops and OFC for a nominal fee of approximately US\$3 (DOA, 1997). This programme helped farmers in the up-country to make substantial savings in the money spent to purchase fertilizers (Maraikar *et al.*, 1996).

Table 12. Application rate of TSP for potato and vegetable crops based on soil test results

Soil P level (mg/kg)	Application rate of the recommended level	Potato	Cabbage	Tomato	Brinjal
<20	Full	270	270	270	325
20-30	Half	200	135	135	165
>30	Starter dose	70	70	60	85

Table 13. Application rate of MOP for potato and vegetable crops based on soil test results (kg/ha)

Soil K level (mg/kg)	Application rate of the recommended level	Potato	Cabbage	Tomato	Brinjal
<160	Full	250	150	130	170
160-400	Half	130	90	90	110
>400	No application	—	—	—	—

20. Pollution of water resources due to agricultural production

Almost one-third of land in Sri Lanka is cropped. Intensive agriculture associated with increase fertilizer and pesticide use has resulted in serious groundwater pollution in some areas of the island. Eastern, northern and some parts of northwestern parts have predominately permeable soils with

shallow water tables, which are more susceptible to leaching with high application rates, making the groundwater potentially hazardous. Groundwater and surface water of central highlands too are becoming polluted with high application of nitrogenous fertilizers from both inorganic and organic sources. The use of fertilizers and pesticides in Sri Lanka is expanding rapidly as in other developing countries, particularly with the introduction of new high yielding crop varieties.

Nagarajah *et al.* (1983) reported that due to intensive cultivation of annual crops in Jaffna peninsula, the nitrate-nitrogen levels exceeds WHO recommended levels. The studies conducted in the Kalpitiya peninsula indicate that leaching of chemical fertilizer from intensively cultivated lands seems to elevate the concentration of nitrates in groundwater (Kuruppuarachchi, 1995). He reported that most irrigation wells had nitrate-N concentrations in excess of the WHO guideline of 11.3 mg N/l and often in excess of 22.6 mg N/l. Further, he reported that build up of nitrate is quite dramatic and has been estimated at 1-2 mg N/l per annum.

In the up-country, potato and vegetables have been cultivated for several decades. The rates of fertilizer applied by farmers are much higher than the quantity recommended by the DOA. The levels of chemical fertilizer applied by farmers to potato and vegetable crop are almost double or triple the quantity recommended by the DOA (Wijewardena, 1996; Wijewardena, 2001). In addition, use of high rates of animal manure is also common in the up-country. Quantities added range from 10 to 15 t/ha of poultry manure and from 20 to 30 t/ha of cattle manure (Wijewardena, 1993; Wijewardena, 1995). Due to the hilly nature and high rainfall in the area, rain could easily wash out applied fertilizer. Wijewardena *et al.* (1995) and Wijewardena *et al.* (1999) monitored drinking water quality of wells in the UCIZ and UCWZ. NO₃-N values reported in these areas were lower than values reported in Jaffna and Kalpitiya areas.

21. Integrated plant nutrition system

During the past decade, average yields of food crops including rice have been stagnant or declining. One of the reasons for this yield decline or stagnation is the unbalanced use of fertilizer. This trend in many cropping systems will result in soil mining leading to decline in soil fertility. In order to improve soil fertility, it is important to follow environmentally-friendly plant nutrition management practices under what has been termed the Integrated Plant Nutrition System (IPNS). This concept advocates the balanced use of fertilizer for crop production. Also, when adding fertilizer to a crop, it would be necessary to assess and take into account the contribution of different sources such as soil, water, organic manure, rain etc. to the nutrient pool.

Studies conducted in relation to plant nutrition management had shown that in many cropping systems the integrated nutrient supply and management through judicious use of organic and chemical fertilizers would lead to sustainable crop production (Tables 14 and 15), as well as overall soil improvement. Hence, the use of this technology is advantageous as it helps to improve fertilizer use efficiency, improvement of long-term soil fertility as well as increase the benefit-cost ratio (Wijewardena and Yapa, 1999; Dissanayake, 2000).

Table 14. Rice yield and VCR with different fertilizer management practices

Treatment	Yield (t/ha)	VCR
Mean yield without chemical fertilizer	3.0	2.68
Mean yield with chemical fertilize	4.5	2.638
Mean yield with chemical fertilize + Rice straw	6.0	5.02
Mean yield with chemical fertilize + Rice straw + Cow dung + Green manure	9.8	8.22
Source: Dissanayake, 2000.		

Table 15. Effect of organic and chemical fertilizers on crop yield (Data is shown from the last three seasons of an eight-season study)

Treatment	Yield (t/ha)		
	Cabbage	Tomato	Bush bean
No fertilizer	6.4	2.2	5.1
NPK	33.0	7.2	10.7
Poultry manure	62.1	18.4	11.1
Poultry manure + NPK	81.8	24.6	15.8
NPK = Rates recommended by the DOA for each crop; Poultry manure = 10 t/ha.			

In general, IPNS is a long-term approach to effectively maintain soil and crop productivity. However, farmers expect immediate and quick economic benefits with IPNS practices. This hinders and restricts adoption of IPNS at the farm level. Thus, creating an awareness of the benefits of IPNS among farmers by introducing various methods of technology transfer would indeed help to solve this problem. In addition, bulky organic manures and its application make IPNS labour intensive. In this respect, Local and Government Organizations should encourage farmers by providing suitable facilities to utilize such materials particularly in rice cultivation, as well as large-scale demonstration, extension services conveying practical training and educational programme to farmers.

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